

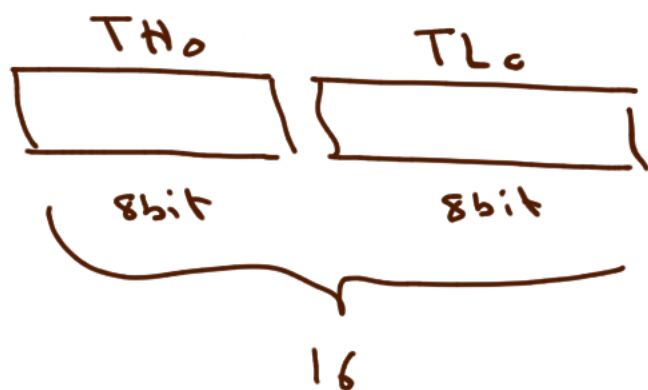
L #9

Timer / counter

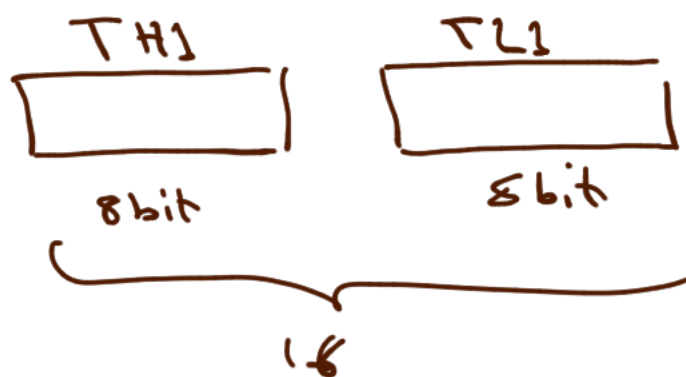
- Is used to generate time delay (time intervals).
- Is used to count outside events.

- 8051 has two timers / counters, Each one is an 16 bit register.

① Timer / counter 0



② Timer / counter 1



MOV TL0, #50H ←

MOV TH0, #0E8H ←



E850H

MOV TL1, R4 ✓

MOV R3, TH1 ✓

* TMOD : Timer mode register (8 bit register) (2)
 is used to configure the mode of timers and counters
 and how the timer/counter operates (start and stop)



M ₁	M ₀	Timer Mode
0	0	0 → use only 13 bit of timer / counter
0	1	<u>1</u> → use 16 bit of " / counter
1	0	<u>2</u> → use only 8 bits of timer / counter
1	1	3 → split mode

auto reload ←

CIT: counter or Timer

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CIT=0 → use it as timer to generate delay.

CIT=1 → " " " counter to count pulses on

P3.4 or P3.5

Gate: is used to define the control type

Gate = 0 → software control

start/stop timer using instructions

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SETB TRx ; start
CLR TRx ; stop
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Gate = 1 → Hardware control

start/stop timer using switch (device)

connected to P3.2 or P3.3

↓
Timer 0

↓
Timer 1

and TR0 and TR1

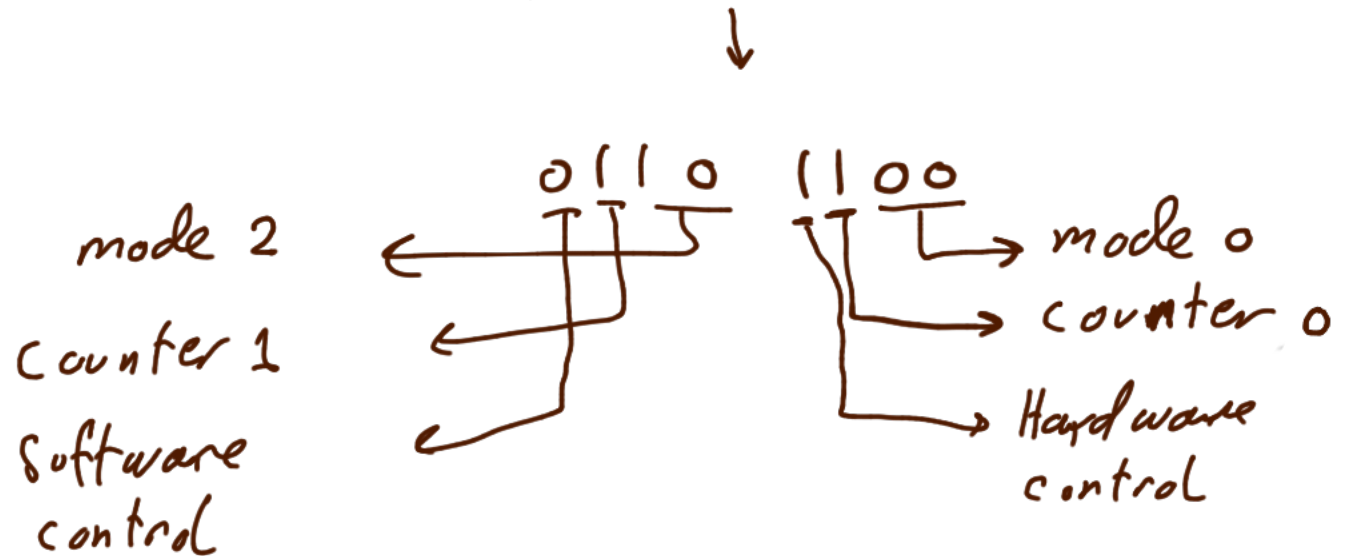
Example: Configure Timer 1 in mode 2 to create time intervals. start/stop timer using P3.3. (4)

→ MOV TMOD, # $\frac{1}{\uparrow} \frac{0}{\uparrow} \frac{1}{\downarrow} \frac{0}{\downarrow} 0000 B$
 → MOV TMOD, # 0A0H

Configure Timer 0 in mode 1 to count outside events and Timer 1 in mode. 2. to generate time delay. start/stop both timers using software control

→ MOV TMOD, # $00 \frac{1}{\downarrow} 0 \frac{0}{\downarrow} \frac{1}{\downarrow} 0 \frac{1}{\downarrow} B$
 → MOV TMOD, # 25H

MOV TMOD, # 6CH



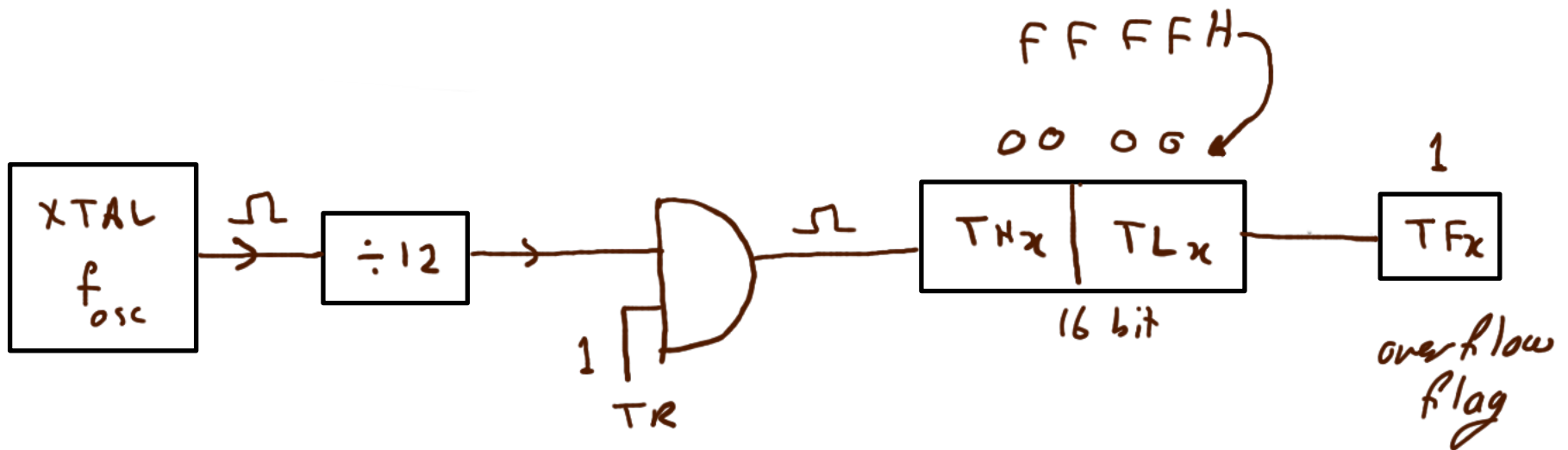
Timer → delay

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① Timer x in mode 1 : use 16 bit

16 bit register (Range of values)

0000H → FFFFH
0 → 65535



max delay → THx, TLx → 0000H

Steps in mode 1 :

① Load TMOD with desired value

MOV TMOD, # —

② load TLx and THx with initial values.

⑥

MOV TLx, # —

MOV THx, # —

③ start timer

SETB TRx

The timer starts count until FFFFH then it rolls over to 0000H then TFx = 1.

④ monitoring TFx

HERE : JNB TFx, HERE

⑤ When TFx = 1 you have to stop timer and clear TFx.

CLR TRx ; stop

CLR TFx ; = 0

⑥ To repeat the delay go to step ②

* calculate values to be loaded in TLX and THX: (7)

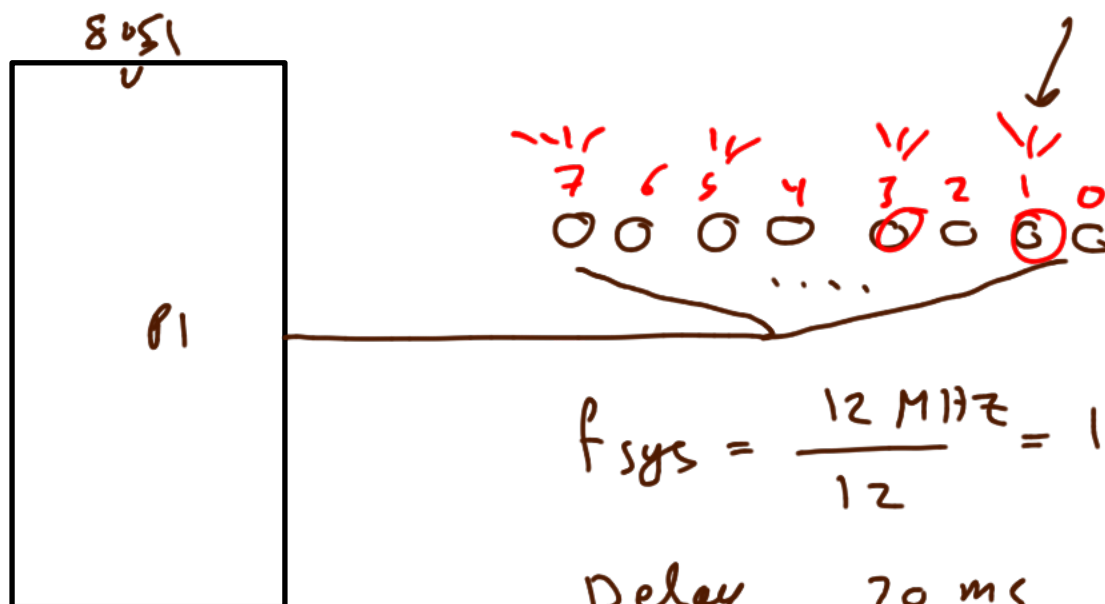
$$(1) f_{sys} = \frac{f_{osc}}{12} \rightarrow MC = \frac{1}{f_{sys}}$$

$$(2) \frac{\text{Delay}}{MC} = \underbrace{\quad}_{\substack{\uparrow \\ \# \text{ of pulses}}} > 256 \rightarrow \text{mode 1}$$

$$(3) 65536 - \quad = K$$

$$(4) \text{convert } K \text{ to Hex} = \begin{array}{cc} YY & LL & H \\ \downarrow & \downarrow & \\ TH & TL & \end{array}$$

write program to turn on/off odd LEDs then even LEDs connected to P1 every 20 ms. Assume $f_{osc} = 12 \text{ MHz}$, use Timer 0.



$$f_{sys} = \frac{12 \text{ MHz}}{12} = 1 \text{ MHz} \rightarrow MC = \frac{1}{1 \text{ MHz}} = 1 \mu s$$

$$\frac{\text{Delay}}{MC} = \frac{20 \text{ ms}}{1 \mu s} = 20000 > 256 \rightarrow \text{mode 1}$$

$$65536 - 20000 = 45536 \\ = B1E0H$$

ORG 0000H

MOV TMOD, #00000001B ^{01H}

MOV A, #01010101B ←

Back: MOV TLo, #0E0H

MOV THo, #0B1H

CPL A

MOV P1, A

SETB TRo

HERE: JNB TFo, HERE

CLR TRo

CLR TFo

SJMP Back

ORG 0000H

MOV TMOD, #01H

Back: MOV P1, #10101010B

ACALL DELAY

MOV P1, #01010101B

ACALL DELAY

SJMP Back

DELAY: MOV TLo, #0E0H

MOV THo, #0B1H

SETB TRo

HERE: JNB TFo, HERE

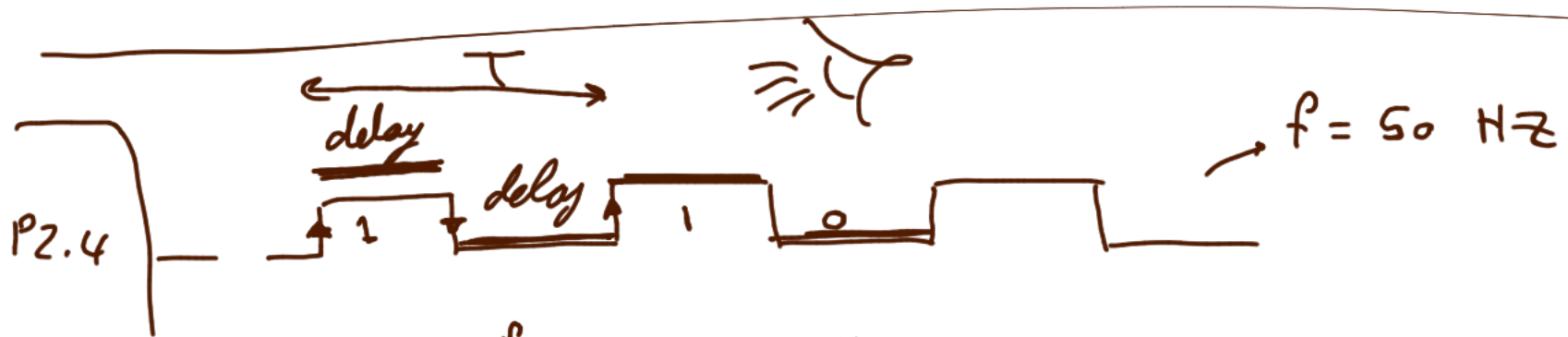
CLR TRo

CLR TFo

RET

} delay 20ms

write a program to create square wave of 50 Hz ⑨
on P2.4. Use Timer 1. $f_{osc} = 24 \text{ MHz}$.



$$f_{sys} = \frac{f_{osc}}{12} = \frac{24 \text{ MHz}}{12} = 2 \text{ MHz}$$

$$MC = \frac{1}{f_{sys}} = \frac{1}{2 \text{ MHz}} = 0.5 \mu s$$

$$\frac{\text{Delay}}{MC} = \frac{?}{MC} = \frac{0.01}{0.5 \mu s} = 20000 > 256 \text{ mode 1}$$

$$T = \frac{1}{f} = \frac{1}{50 \text{ Hz}} = 0.02 \text{ sec}$$

$$\text{delay} = \frac{T}{2} = \frac{0.02}{2} = 0.01 \text{ sec}$$

$$65536 - 20000 = 45536 \rightarrow \text{BIEGH}$$

ORG 0

MOV TMOD, # 0001 0000B

Back : MOV TL1, # 0E0H

MOV TH1, # 0B1H

SETB TR1

HERE : JNB TF1, HERE

CPL P2.4

CLR TR1

CLR TF1

SJMP Back

ORG 0

MOV TMOD, # 0001 0000B

Back : SETB P2.4

ACALL DELAY

CLR P2.4

ACALL DELAY

SJMP Back

Back : CPL P2.4

ACALL DELAY

SJMP Back

DELAY: MOV TL1, # —

MOV TH1, # —

SETB TR1

H1 : JNB TF1, H2

CLR TR1

CLR TF1

RET